

ELOQUENT CLEF Shared Tasks for Evaluation of Generative Language Model Quality, 2025 edition

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Abstract. The ELOQUENT lab for evaluation of generative language model quality and usefulness addresses high-level quality criteria through a set of open-ended shared tasks implemented, where possible, to leverage the ability of systems built on language model to assess their own capacity.

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1 Assessing the Quality of Generative Language Models

Generative language models (“LLMs”) have become a foundational technology for both research and information service development, and have prompted great public interest through some quite innovative application scenarios.

A generative language model as a component in an information service can handle a broad variety of input data robustly and elegantly, and can provide appropriately creative generated output to fit the varied application situations and the preferences of a diverse user population.

The ELOQUENT evaluation lab experiments with new evaluation methods for generative language models to meet some of the challenges in the path from laboratory to application. Deploying generative models, or selecting among the several recent models with various generative capacities, is challenged by runtime costs, training costs, and to a great extent by concerns about the quality, predictability, appropriateness, and reliability of the output from the generative component. In research contexts, quality assurance, evaluation, and benchmarking of large language models is performed using tests that focus on fluency, details

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about language understanding and some aspects of reasoning using language. There is a large number of tasks available for this purpose. While a language model can provide a system with fluently generated output a language model issues no guarantees of relevance, veracity, or consistency, factors that are crucial for application development and deployment to third parties. These factors not addressed systematically through existing benchmarks. In addition, the current benchmarks are developed almost exclusively for English, which raises questions about the quality of output in other languages, or in multilingual settings.

The ELOQUENT lab develops tasks to establish useful top-level quality criteria such as resilience, robustness, and consistency, correlation with human preferences and cultural norms. and topical sensemaking. ELOQUENT tasks are intended to involve comparatively little human effort. We instead leverage the capacity of the current generation of AI models to generate, process, and assess human language input. This is a direction which lately has garnered great interest, with obvious risks and vulnerabilities [6,17,1,3] and we intend address some of the challenges by using participant systems as judges rather than using a single oracle, to broaden the response range of the automated assessments by combining cross-model evaluation where one model evaluates another’s output [10,14], and by self-evaluation where the model evaluates its own output [10,11].

2 Learnings from the First Edition and Plans for the Second Edition

In its first edition, ELOQUENT organised four tasks to investigate *topical competence, truthfulness and trustworthiness, robustness* in face of varied input, and the difference between *human-authored vs machine-generated* text. More details of each are given below, and a more comprehensive overview has been published in the CLEF proceedings [7].

ELOQUENT received great general interest, and 55 registration sign-ups for teams to participate in various subsets of the four tasks. Of these, 8 teams submitted experiment results. This is a high attrition rate, and polling participants and non-participating registrants found that an earlier timeline and a less explorative and more strict exploratory approach would be approached. These lessons were taken to heart and this coming edition we will release task details and sample data earlier, and have the ambition of formulating tasks on either on two levels of ambition: a research-oriented open-ended exploratory experiment or a more constrained and scoreable lab assignment.

Of the four tasks from the first edition, two tasks will continue into the next edition with small changes, one will be reformulated, and one will move to another venue; one new tasks will be introduced.

Voight-Kampff Task (Continuing from 2024)

Recent advances in generative language models have made it possible to automatically generate content for websites, news articles, social media, and other

channels. There are calls to label machine generated content as such, motivated e.g. by consumer protection, liability concerns, copyright issues, as well as many further reasons. This challenge is in some obvious ways quite similar to human authorship verification, and the *Voight-Kampff Task* is part of a joint effort with the PAN lab at CLEF to address that challenge, bringing together years of experience from previous shared tasks organised by the PAN lab on authorship verification and closely related tasks with the focus on generative artificial intelligence that ELOQUENT participants have from their work on model building, training, tuning, and evaluation. Inspired by Ettinger et al [5], the Voight-Kampff participants can elect to participate in either or both of two roles: *builder*, to design and implement systems to discriminate between automatically-generated and human-authored texts, or *breaker*, to generate test data for a dataset of parallel human-authored and automatically generated language to be submitted to the builders' systems to challenge their discriminative capacity. The objective of the builders is to be able to distinguish the two types of text; the objective of the breakers is to fool the builders' systems.

This task ran in the first edition and will continue this year, with details, participants, and results given in the overview paper[2]. In the first edition, the submitted samples were very handily identified by the participating classifiers, and the challenge to provide convincing human-like output remains unmet. The task is intended to have a low threshold for participation, and we hope that all or most participants in the other tasks will also submit test data for this task.

Hallucigen Task (Moving elsewhere for 2025)

Generative language models will output language that adheres to the training data the model has been exposed to. Blending that previously seen material into a coherent output is no guarantee that the results adheres to human experiences of the world. The *HalluciGen task* addressed the challenge to identify when a language model generates output which diverges from human reality, or "hallucinates". The experiments focussed on two scenarios: one where a model paraphrases some given text item, the other where a model translates text from one language to another. The participating models were requested to generate test data and to process data, thus providing support both for cross-model evaluation and self-evaluation tests.

Resulting datasets are available on Huggingface.⁷ More detail on the task, the participants, and on the results are given in the overview paper for the task [4]. This task is planned to move to another venue in 2025.

Robustness and Consistency Task (Continuing from 2024)

Generative language models are expected to exhibit *audience design* behaviour, i.e. to fit their output to the preceding input. In general, this is desirable and

⁷ <https://huggingface.co/datasets/Eloquent/HalluciGen-PG>
<https://huggingface.co/datasets/Eloquent/HalluciGen-Translation>

emulates important aspects of human linguistic behaviour. However, if this variation extends to content-related aspects of the output, this may have unfortunate systematical effects if a system is sensitive to dialectal, sociolectal, cross-cultural variation and uses that understanding to provide different material to different demographic groups to the extent that they receive e.g. different medical, financial, or legal information.

This task assesses the how elicited responses from generative language models vary by input prompt variation. The task provides participants with a set of input prompts that vary along some dimensions of style and form to be submitted to systems. The resulting outputs from participating systems will be scored both through self-evaluation and cross-model assessment for some observable content-related variables of interest. This task has a natural multi-lingual component, and the first edition provided multi-lingual variants of the prompts. This second edition, the multi-lingual aspect will be emphasised further. The first edition provided some examples of how using the word "college" pro "university" in the prompt steered output to a US-based perspective and how a prompt which used medical terminology rather than everyday language steered output from general advice to specifics, even when the terms were ill chosen. [13,15,12]

Preference Prediction Task (New for 2025)

Side-by-side evaluation has become a well-established paradigm for assessing how well language models align with human preferences across various tasks. Recent research has explored methods to automatically evaluate the language model alignment using oracle models to mitigate the cost of collecting human-based preference data [19,9]. However, the ability of language models to reproduce human preferences and justify their choices are still underexplored [16,18].

The Preference Prediction Task is a new task for this second edition, and it addresses the development of systems capable of predicting side-by-side human preferences and explaining their choices with respect to standard evaluative criteria, ranging from relevance to truthfulness. We will provide the participants with a novel dataset of human-based preferences and judgments, which will be collected from scratch. This task offers two sub-tasks with participation open to anyone: (1) *Preference prediction*. The task is to predict a coarse-grained preference between two given language model responses and fine-grained preferences across a set of evaluative criteria. The performance metric is the accuracy score. (2) *Preference prediction and judgment generation*. In addition to predicting coarse- and fine-grained preferences, a system is required to generate judgements explaining the choices. The performance metrics include standard natural language generation evaluation metrics. Moreover, we will conduct an in-house human-based evaluation of top-k systems.

All human-based data collection and evaluation will be conducted on Toloka⁸, a data labelling platform.

⁸ toloka.ai

Sensemaking Task (An Evolution of the 2024 Topical Quiz Task)

A generative language model in practical application will in most envisioned use cases be expected to stay within topical boundaries, to generate material restricted to the domain it is employed to work within, and to have competence in the terminology and conventions of that domain. To this end, in the first edition of ELOQUENT, the *Topical Quiz Task* was formulated to answer to the need for verifying a model’s understanding of an application domain of interest. The task was defined for a model to generate a topical quiz for a series of topics, respond to its own quiz and to other quizzes generated by other systems, and to score their own and others’ responses [8,12].

The topical quiz task has been for this second edition of ELOQUENT further developed into the *Sensemaking Task* where the participating systems will be provided with a limited amount of texts to use as a base to generate a similar quiz as above, allowing the experiment to control better for topical variables. The focus this year will be not only on staying on topic, but also *staying within the content* of the provided material. A relevant real-world example of this situation are school exams where quiz questions would be unfair if they ask about something which was not covered in study materials; students taking the quiz are not penalised for knowing extra detail but they would be for not knowing information from the materials. The foreseen evaluation in the Sensemaking Task will distinguish between answers that are correct and complete *according to the materials*, or which rely on other knowledge. The task focus is on the first case.

We anticipate participants at any of the following levels: (1) a “teacher” prepares a system that, given input materials, generate quizzes and score answers; (2) a “student” prepares a system that, given input materials and quiz questions, answers the questions; (3) a “director” provides input materials and is required to analyse what “teachers” and “students” did in that domain.

The datasets currently considered for Sensemaking Task include multi-modal material, both in original form and transcribed to plain text, material in several languages, and material from several genres (primary school textbooks, news material, university level teaching materials).

Experimental Objectives

The ELOQUENT evaluation lab intends to provide support for developing self-assessment methodologies for large language models and to investigate the reliability of using models to judge the quality of other models. We expect the tasks to evolve during the year: next year’s tasks will incorporate learnings from this year to establish where a language model is to be trusted and where it needs language-external functionality to be a useful component in an information system.

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